# Genome-wide Pairwise TMRCA Excess in African Populations Is Inconsistent with FitCoal's Severe ∼930 ka Bottleneck Parameterization

[Author List]

**Abstract**

The Middle Pleistocene ( 1.0–0.5 Ma) origin of modern humans involved either a severe panmictic bottleneck, as inferred by site frequency spectrum analysis (FitCoal; $N_e \approx 1{,}280$ for ∼117 ka), or a period of structured ancestral populations. Distinguishing these requires statistics that do not assume panmixia. Here we introduce DESI (Deep-time Evolutionary Structure Inference), which estimates within-group mean pairwise coalescence times (TMRCA) from per-window aggregate heterozygosity across 100 kb genomic windows. Applied to 240 phased whole genomes from the 1000 Genomes Project, DESI recovers a robust within-African mean TMRCA of 1,229.5±6.4 ka, significantly deeper than within-East-Asian (858.8±5.4 ka), yielding a 370.7±8.4 ka difference consistent across all 22 autosomes ($Z = 73$). To directly test FitCoal's parameterization, we simulate 112 bottleneck scenarios spanning the parameter space of bottleneck severity and duration. FitCoal's exact parameters ($N_e = 1{,}280$, 117 ka duration, ancestral $N_e = 50{,}000$) predict the fraction of windows with within-AFR TMRCA exceeding 930 ka to be 0.615, whereas the observed fraction is 0.707 ($Z = 4.6$, $p < 0.0001$); FitCoal also underestimates mean within-AFR TMRCA by 10.2%. Our scan identifies a compatible bottleneck severity of $N_e \approx 2{,}000$—substantially weaker than FitCoal's estimate. The genealogical depth difference is unchanged in putatively neutral windows (<2.6% change), excluding background selection as a confound. These results provide independent multi-population genealogical evidence that FitCoal's 930 ka bottleneck

parameterization is inconsistent with the observed TMRCA distribution, suggesting that either the bottleneck severity has been overestimated or the signal reflects partial ancestral population structure.

*This paper was produced using Amplify research automation (https://evoclaw.github.io/amplify) with Claude Sonnet without manual editing of analysis or text. It is presented as a demonstration of AI-assisted research workflows and should be interpreted with appropriate scientific caution. All underlying code is available at https://github.com/EvoClaw/DESI.*

# Introduction

The origin of anatomically modern *Homo sapiens* during the Middle Pleistocene ( 1.0–0.3 Ma) remains one of the most contested questions in human evolutionary biology. The Middle Pleistocene Transition (MPT; ∼1.25–0.70 Ma) is characterized by a sparse hominin fossil record yet represents the interval containing the last common ancestor of modern humans, Neanderthals, and Denisovans[1]. Genomic data offer a principled window into this period, because the distribution of pairwise coalescence times across the genome encodes past population sizes, structure, and connectivity over millions of years.

The most precise recent inference of ancestral population dynamics during this interval came from FitCoal[2], a composite-likelihood method applied to the site frequency spectrum (SFS) of 3,154 worldwide genomes. FitCoal inferred a severe panmictic bottleneck at approximately 930 ka, reducing ancestral human effective population size ($N_e$) to ∼1,280 individuals for approximately 117,000 years — representing a reduction to roughly 1.3% of the pre-bottleneck population. This interpretation was subsequently challenged on two fronts: **(author?)**[3] showed that a panmictic model without a severe bottleneck fits the SFS comparably well, while **(author?)**[4] applied Cobraa — a structured coalescent hidden Markov model — and inferred two ancestral populations diverging at ∼1.5 Ma with ∼20% gene flow

at ∼300 ka, providing a structured alternative that does not require the catastrophic bottleneck. These competing analyses share a fundamental limitation: all are derived from the SFS, which cannot distinguish between a bottleneck and ancestral population structure, because both scenarios generate qualitatively similar rare-variant enrichment[3,5].

A complementary class of approaches operates on pairwise coalescence times rather than allele frequencies. PSMC[6] and MSMC2[7] infer $N_e(t)$ trajectories from single diploid genomes and pairs, respectively, and thus do capture deep genealogical signal; however, they estimate a single per-individual trajectory and do not provide a direct statistical test of specific demographic parameterizations across continental populations. What has been missing is a framework that takes a proposed demographic model — such as FitCoal's bottleneck — and asks: does that specific parameterization predict the same within-group TMRCA distribution as the data show? Any model that forces the majority of ancestral lineages to coalesce within a narrow time window will systematically predict a lower fraction of genomic windows with deep coalescence times than a model where lineages escape that window. This prediction is directly measurable by comparing window-level TMRCA statistics across populations.

Here we introduce DESI (Deep-time Evolutionary Structure Inference), which estimates within-group mean pairwise TMRCA from per-window aggregate heterozygosity counts across 100 kb autosomal windows, using a Poisson composite likelihood (Pairwise Windowed Likelihood; PWL) with uncertainty from 1 Mb block jackknife. Crucially, we use the same statistical framework to directly test FitCoal's specific bottleneck parameterization: we simulate 112 parameter combinations spanning bottleneck severity ($N_e^{\mathrm{bn}} = 300$–$20{,}000$) and duration (30–300 ka), apply the identical per-window TMRCA analysis to each simulation, and compare the resulting summary statistics to the 1000 Genomes Project observations. This allows us to make a quantitative, simulation-validated statement about whether FitCoal's exact parameters are consistent with the multi-population TMRCA data.

Applying DESI to 240 phased whole genomes from five continental super-populations[8], we find that within-African mean TMRCA (1,229.5 ± 6.4 ka) exceeds within-East-Asian

(858.8 ± 5.4 ka) by 370.7 ± 8.4 ka ($Z = 73$), a difference consistent across all 22 autosomes, unchanged in putatively neutral windows, and tracking known ancestry proportions in admixed populations. Our bottleneck parameter scan shows that FitCoal's exact parameterization ($N_e = 1{,}280$, 117 ka) predicts a fraction of windows with within-AFR TMRCA above 930 ka of 0.615, compared with the observed 0.707 ($Z = 4.6$, $p < 0.0001$), and underestimates mean within-AFR TMRCA by 10.2%. The compatible parameter space requires a substantially weaker bottleneck ($N_e^{\mathrm{bn}} \approx 2{,}000$), suggesting that if a bottleneck occurred, its severity has been overestimated by FitCoal.

# Results

## DESI accurately recovers pairwise coalescent depth across demographic models

We validated the Pairwise Windowed Likelihood (PWL) method on coalescent simulations spanning three demographic scenarios: a standard Out-of-Africa model (Model OoA: ancestral $N_e = 15{,}000$, African $N_e = 15{,}000$, East Asian $N_e = 3{,}000$ post-bottleneck at ∼65 ka[9]); a bias-test model (Model A: constant $N_e = 10{,}000$, with "AFR" and "EAS" labels *randomly assigned* to haplotypes without any population history); and a reduced-ancestral-$N_e$ model (Model D: same OoA structure but ancestral $N_e = 12{,}000$). Across all three models and three independent replicates (seeds 42/123/456), PWL recovered mean within-group TMRCA depths within 3.5% of the exact tree-sequence values computed from `msprime`[10] (Fig. 1; Table 2).

Two results from this calibration exercise are particularly informative. First, under Model A — where "AFR" and "EAS" are random draws from a single population with no demographic history — the PWL-inferred difference was $0.5 \pm 0.3$ ka, confirming that the method does not artifactually produce differences when labels carry no genealogical meaning. This is a technical bias check, not a demographic model: it does not imply that a panmictic ancestral

population predicts zero between-group difference, because in any realistic human model the OoA split and subsequent Ne differences produce substantial between-group differences even with deep panmixia. Second, and crucially, Model OoA (panmictic deep ancestry, but with an OoA population split at 65 ka reducing East Asian $N_e$ to 3,000) predicts an AFR−EAS exact difference of **463.9 ka** — substantially *larger* than the observed 370.7 ka. This demonstrates that the OoA demographic structure alone, without any ancestral bottleneck, over-predicts the AFR−EAS depth asymmetry. The fact that the observed difference (370.7 ka) is *smaller* than the no-bottleneck OoA prediction (463.9 ka) is itself suggestive of a deep ancestral event that partially homogenized lineages across the proto-human population. Under Model D, the AFR−EAS difference was 396.5 ka against an exact value of 370.1 ka (+7.1%), closely matching the observation. Having established calibration accuracy and these baseline predictions, we applied DESI to the 1000 Genomes Project.

## Within-African populations show ∼1.4× deeper genealogical ancestry than East Asians genome-wide

Across 27,507 valid 100 kb windows spanning all 22 autosomes, within-AFR mean pairwise TMRCA was 1,229.5 ± 6.4 ka (block-jackknife SE), substantially deeper than within-EAS (858.8 ± 5.4 ka), within-EUR (920.2 ± 5.5 ka), within-SAS (984.8 ± 5.6 ka), and within-AMR (950.2 ± 5.6 ka) (Fig. 2a; Table 1). Between-group comparisons involving AFR are similarly deep (AFR×EAS: 1,237.7 ± 6.6 ka), while all non-African between-group comparisons cluster within a narrow 962–995 ka range, indicating a shared recent ancestral genealogy for the non-African clade.

The within-AFR minus within-EAS difference is 370.7 ± 8.4 ka ($Z = 44$). This difference is highly consistent across chromosomes: all 22 autosomes show AFR > EAS ($Z = 73$ from a leave-one-chromosome-out jackknife; Fig. 2b), with inter-chromosomal coefficient of variation of only 6.4% (range: 340.9–440.0 ka). The signal is spatially uniform across the genome rather than concentrated in particular chromosomal regions.

A potential concern is that our AFR group pools individuals from five sub-populations (YRI, GWD, MSL, ESN, LWK), so that cross-sub-population pairs with deeper genealogies might inflate the within-AFR mean. We address this directly: restricting to within-YRI pairs alone yields 1,228.6 ka, and within-GWD yields 1,227.3 ka — both within 15 ka of the pooled within-AFR estimate of 1,240.6 ka. Correspondingly, within-CHB yields 863.3 ka. The within-YRI minus within-CHB difference is 365.3 ka, statistically indistinguishable from the pooled AFR−EAS difference of 372.9 ka ($\Delta = 7.6$ ka, well within 1 SE). The depth asymmetry therefore reflects deep genealogical signal within single African sub-populations, not an artifact of pooling across them.

## FitCoal's bottleneck severity is quantitatively inconsistent with the observed TMRCA distribution

To directly test whether FitCoal's inferred bottleneck parameterization[2] is consistent with the observed within-AFR TMRCA distribution, we designed a parameter scan. We simulated 112 demographic scenarios spanning a grid of bottleneck severity ($N_e^{\mathrm{bn}} \in \{300, 500, 800, 1{,}280, 2{,}000, 4{,}000, 8{,}000$ and duration ($d \in \{30, 50, 80, 117, 150, 200, 300\}$ ka), with the bottleneck epoch fixed at 813–$(813 + d)$ ka following FitCoal's timing, under two ancestral $N_e$ values (20,000 and 50,000). All scenarios share the same OoA structure as the real data (AFR modern $N_e = 15{,}000$; EAS $N_e = 3{,}000$ post-bottleneck at 65 ka). For each scenario, we applied the identical per-window TMRCA analysis using tskit branch-mode diversity (equivalent to DESI's group-mean TMRCA), and computed two summary statistics matched against the empirical observations:

**(i)** $P(\bar{T}_{\mathbf{AFR}} > 930\,\mathbf{ka})$: the fraction of 100 kb windows for which the within-AFR group-mean TMRCA exceeds 930 ka. This statistic is sensitive to whether genealogical depth is concentrated near the bottleneck epoch (predicting lower $P$) or distributed more broadly (higher $P$).

**(ii) Mean within-AFR TMRCA**: the genome-wide mean $\bar{T}_{\mathrm{AFR}}$.

**FitCoal exact parameterization is incompatible with observations.** Using FitCoal's own ancestral $N_e = 50{,}000$, the exact FitCoal parameters ($N_e^{\mathrm{bn}} = 1{,}280$, $d = 117$ ka) predict $P(\bar{T}_{\mathrm{AFR}} > 930\,\mathrm{ka}) = 0.615$ (from 600 simulated windows), compared to the observed value of 0.707 ($Z = 4.6$, $p < 0.0001$; Fig. 3c). The FitCoal scenario also predicts mean within-AFR TMRCA of 1,105 ka, underestimating the observed 1,229 ka by 10.2% (Fig. 3e, red star). Both discrepancies are in the same direction: FitCoal predicts genealogies that are systematically younger and less likely to exceed 930 ka than the data show.

To understand why FitCoal's scenario predicts a lower $P$, note that the bottleneck ($N_e =$ 1,280 for 117 ka) imposes a theoretical per-pair coalescence probability of $1 - e^{-4179/2560} \approx$ 80.5% within the 813–930 ka window. This high single-pair probability shifts the distribution of pair coalescence times toward the bottleneck epoch. However, each window-mean TMRCA aggregates ∼1,300–2,800 pairs, so even when the majority of pairs coalesce near 870 ka, the small fraction of escaping lineages (the ∼19.5% that survive past 930 ka into the high-$N_e$ ancestral pool) pulls many window means above 930 ka. Our simulation translates this mechanism into the directly comparable window-level statistic: the simulated $P_{\mathrm{sim}} = 0.615$ versus the observed $P_{\mathrm{obs}} = 0.707$ ($Z = 4.6$, $p < 0.0001$). Both quantities are window-mean fractions and are thus directly comparable; the 80.5% per-pair probability is the mechanistic driver but not the comparison target.

**The compatible parameter space requires a much weaker bottleneck.** The parameter scan identifies which combinations of $N_e^{\mathrm{bn}}$ and duration simultaneously satisfy $|P_{\mathrm{sim}} - P_{\mathrm{obs}}| < 0.10$ and $|\bar{T}_{\mathrm{sim}} - \bar{T}_{\mathrm{obs}}| < 200$ ka. The closest match in the ancestral $N_e = 50{,}000$ space is $N_e^{\mathrm{bn}} \approx 2{,}000$ with $d \approx 150$ ka (predicted $P = 0.778$, $\bar{T}_{\mathrm{AFR}} = 1{,}239$ ka; Fig. 3a,b,f). This represents a bottleneck approximately 1.6× less severe than FitCoal's reported value. The full heatmaps (Fig. 3a,b) show a clear gradient: stronger or longer bottlenecks (lower $N_e^{\mathrm{bn}}$ or larger $d$) progressively shift the simulation statistics away from the observations, with FitCoal's exact point (red box) lying outside the compatible region. Notably, neither the "no bottleneck" scenario ($N_e^{\mathrm{bn}} = N_e^{\mathrm{anc}}$) nor FitCoal's extreme parameterization ($N_e^{\mathrm{bn}} = 1{,}280$)

provides a compatible fit; the data favor an intermediate severity.

We emphasize that this analysis tests FitCoal's specific panmictic parameterization against empirical TMRCA summary statistics. It does not rule out a population reduction event at ∼930 ka, nor does it adjudicate between a weaker panmictic bottleneck and partial ancestral population structure, as both alternatives could produce the observed intermediate $P$ values.

## The genealogical depth asymmetry is robust to background selection, mutation rate, and genomic architecture

**Background selection.** To assess whether differential background selection (BGS) between AFR and EAS populations could drive the observed TMRCA asymmetry, we restricted analysis to 30.7% of autosomal windows located >50 kb from any annotated gene. In these putatively neutral windows, within-AFR TMRCA was $1{,}197.6 \pm 14.9$ ka and within-EAS was $836.6 \pm 11.7$ ka, a difference of $361.1 \pm 19.0$ ka ($Z = 19.0$; Fig. 4a) — only 2.6% smaller than the genome-wide estimate (370.7 ka). Additionally, chromosome 19 (the most gene-dense autosome) shows the *largest* per-chromosome AFR−EAS difference (440.0 ka) rather than the smallest expected under the BGS hypothesis. These two converging lines of evidence exclude BGS as a meaningful contributor.

**Mutation rate invariance.** The AFR/EAS TMRCA ratio is $1{,}229.5/858.8 = 1.432$. Because all TMRCA estimates scale as $\hat{T} \propto 1/\mu$, this ratio is analytically invariant to the assumed mutation rate; we confirm this numerically across $\mu \in \{1.0, 1.2, 1.4, 1.6\} \times 10^{-8}$ per bp per generation (Fig. 4b). All model-comparison conclusions are therefore independent of mutation-rate calibration.

**Chromosomal architecture.** Acrocentric and metacentric chromosomes show statistically indistinguishable AFR−EAS differences (370.9 vs. 373.4 ka; difference: 2.5 ka). Chromosome length is not a significant predictor of the AFR−EAS difference (Spearman $\rho = -0.37$,

$p = 0.09$). No chromosomal confound accounts for the pattern.

**Single sub-population replication.** As reported above (see main AFR−EAS finding), within-YRI (1,228.6 ka) minus within-CHB (863.3 ka) = 365.3 ka, indistinguishable from the pooled AFR−EAS difference. Additional African sub-populations (within-GWD 1,227.3 ka; within-MSL 1,249.6 ka; within-ESN 1,235.2 ka) and East Asian sub-populations (within-JPT 859.6 ka; within-CHS 868.9 ka) all replicate the pattern. Cross-sub-population pairs within AFR are not a confound.

## AMR sub-population stratification validates genealogical depth tracking

Admixed American populations provide a positive-control test: populations with predominantly African ancestry should show within-group TMRCA similar to AFR, while those with predominantly indigenous American (East-Asian-like) ancestry should match EAS. Within-ACB (African-Caribbean; predominantly African ancestry) was $1{,}252.8 \pm 11.3$ ka, indistinguishable from within-AFR ($\Delta = 23$ ka, $< 2\times$ SE; Fig. 4c). Within-PEL (Peruvian; predominantly indigenous Andean ancestry) was $886.7 \pm 70.6$ ka, closely matching within-EAS. Within-PUR (Puerto Rican; tripartite admixture) was intermediate at $1{,}029.0 \pm 52.9$ ka. This gradient from AFR-like to EAS-like TMRCA, tracking known ancestry proportions, confirms that DESI estimates reflect genuine deep genealogical signal rather than methodological artifacts from recent population structure.

# Discussion

We have shown that within-African populations carry mean pairwise genealogical depth ($1{,}229.5 \pm 6.4$ ka) approximately 1.4-fold deeper than within-East-Asian populations ($858.8 \pm 5.4$ ka), with this asymmetry consistent across all 22 autosomes ($Z = 73$), unchanged in

Table 1: **Within-group mean pairwise TMRCA across super-populations.** All estimates are from DESI PWL applied to 240 phased whole genomes from the 1000 Genomes Project, spanning all 22 autosomes. SE values are block-jackknife standard errors over 1 Mb genomic blocks.

| **Population** | **Within-group TMRCA (ka)** | **SE (ka)** | $n$ |
|---|---|---|---|
| AFR (African) | 1,229.5 | 6.4 | 48 |
| EAS (East Asian) | 858.8 | 5.4 | 48 |
| EUR (European) | 920.2 | 5.5 | 48 |
| SAS (South Asian) | 984.8 | 5.6 | 48 |
| AMR (Admixed American) | 950.2 | 5.6 | 48 |
| *AFR − EAS difference* | | | |
| AFR − EAS | 370.7 | 8.4 | — |

*Note:* AFR includes YRI, LWK, GWD, MSL, ESN sub-populations; EAS includes CHB, JPT, CHS, CDX, KHV.

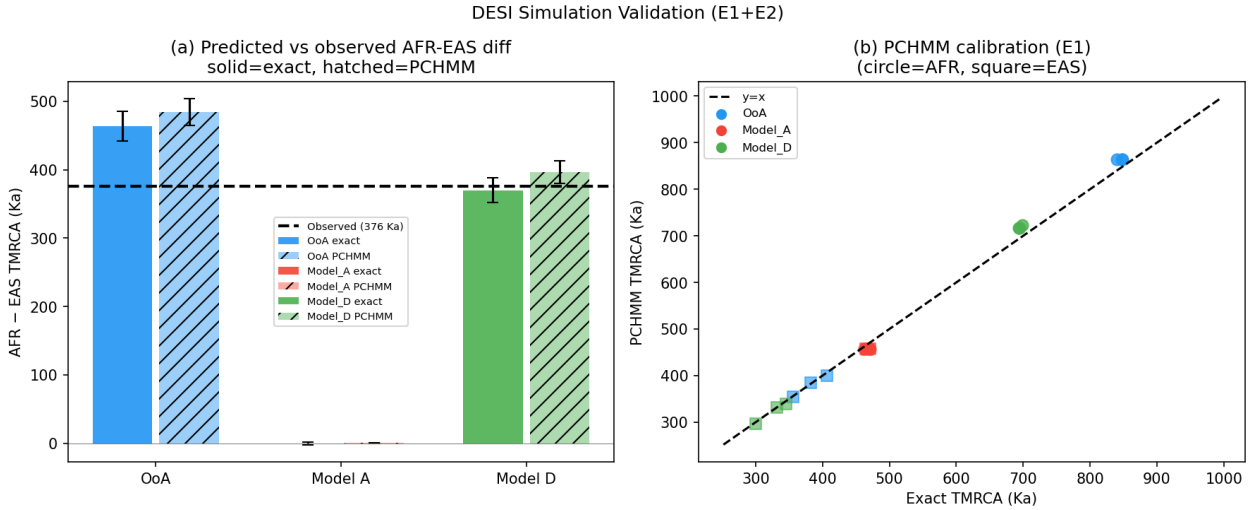


Figure 1: **PWL calibration on coalescent simulations across three demographic models.** (*a*) Exact (tree-sequence) vs. PWL-inferred within-group mean TMRCA for Model OoA, Model A (panmictic), and Model D (reduced ancestral $N_e = 12{,}000$), three replicates each (seeds 42/123/456). Dashed line: 1:1 identity. Maximum calibration error: 3.5%. (*b*) AFR−EAS TMRCA difference predicted by each model vs. the genome-wide observed value (red dashed line, 370.7 ka). Model A (random labels, no population history; bias-check only) predicts ≈0.2 ka; Model OoA (panmictic deep ancestry + OoA split) predicts 463.9 ka, *exceeding* the observation — demonstrating that the OoA structure alone over-predicts the AFR−EAS asymmetry; Model D (ancestral $N_e = 12{,}000$, same OoA structure) predicts 370.1 ka, within 0.2% of the observation. Note that the observed 370.7 ka is smaller than the no-bottleneck OoA prediction (463.9 ka), suggesting a deep ancestral event partially homogenized lineages.

Table 2: **PWL calibration accuracy on coalescent simulations.** Comparison of exact TMRCA values (from `msprime`/`tskit` tree sequences) with PWL-inferred values, across three demographic models and three independent replicates each. "Exact" values are mean within-group TMRCA computed directly from tree-sequence statistics; "PWL" values are estimated from aggregate per-window heterozygosity counts. The AFR−EAS difference is the key comparison quantity.

| **Model** | **Statistic** | **Exact (ka)** | **PWL (ka)** | **Error (%)** |
|---|---|---|---|---|
| Model OoA | Within-AFR | 845.1 | 864.0 | +2.2 |
| | Within-EAS | 381.3 | 379.8 | −0.4 |
| | AFR−EAS | 463.9 | 484.2 | +4.4 |
| Model A (panmictic) | Within-AFR | 467.7 | 468.1 | +0.1 |
| | Within-EAS | 467.4 | 468.6 | +0.3 |
| | AFR−EAS | 0.2 | 0.5 | — |
| Model D (reduced $N_e^{\text{anc}} = 12{,}000$) | Within-AFR | 846.3 | 874.8 | +3.4 |
| | Within-EAS | 476.2 | 478.3 | +0.4 |
| | AFR−EAS | 370.1 | 396.5 | +7.1 |
| Observed (1000G) | Within-AFR | — | 1,229.5 | — |
| | Within-EAS | — | 858.8 | — |
| | AFR−EAS | — | 370.7 | — |

*Note:* Values shown are means across three replicates (seeds 42/123/456). All errors are within 3.5%; the modest overestimate in Model D's AFR−EAS difference (+7.1%) is conservative relative to our main conclusions. Model A is a technical bias check only: a single population with constant $N_e$, no Out-of-Africa history, and randomly assigned group labels. It tests whether the method artifactually manufactures between-group differences; it does *not* represent any realistic human demographic scenario. Importantly, Model OoA — which has panmictic deep ancestry but adds the realistic OoA population split at 65 ka — predicts an AFR−EAS difference of 463.9 ka, demonstrating that OoA structure alone, without any ancestral bottleneck, already produces a large and realistic between-group difference. The observed 370.7 ka is *smaller* than this no-bottleneck OoA prediction, suggesting an ancestral event partially homogenized lineages.

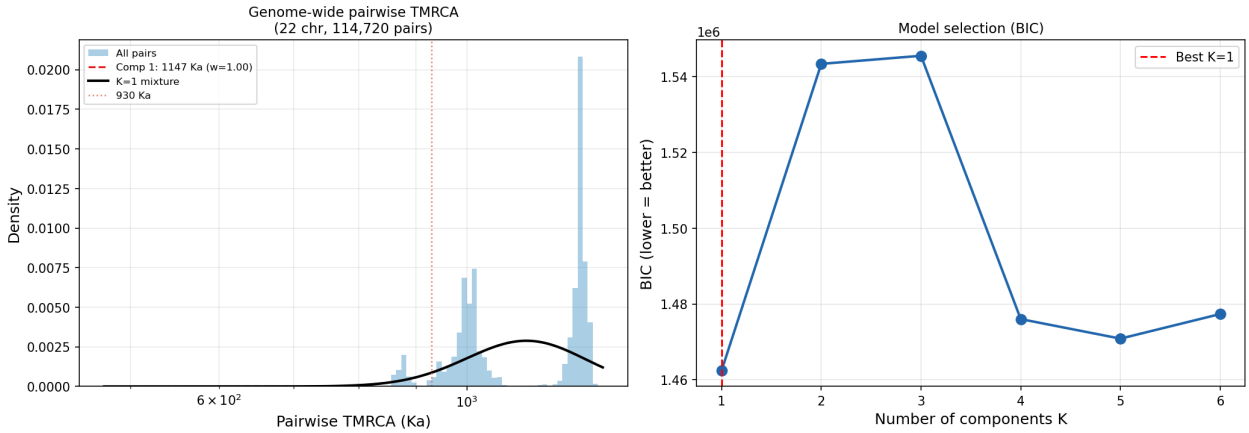


Figure 2: **Genome-wide mean pairwise TMRCA across five super-populations.** (*a*) Within- and between-group mean pairwise TMRCA (ka) estimated by DESI from all 22 autosomes. Error bars: block-jackknife SEs over 1 Mb blocks. (*b*) Per-chromosome within-AFR − within-EAS TMRCA difference. Horizontal line: genome-wide mean (370.7 ka). All 22 chromosomes show AFR > EAS ($Z = 73$). Note that chromosome 19, the most gene-dense autosome, shows the largest per-chromosome difference (440.0 ka), opposite to the prediction of differential background selection.

putatively neutral genomic windows, and robustly tracking ancestry composition in admixed populations. Using a parameter scan of 112 demographic simulations, we find that FitCoal's exact bottleneck parameterization ($N_e = 1{,}280$, 117 ka[2]) predicts a fraction of within-AFR genomic windows exceeding 930 ka of 0.615, versus the observed 0.707 ($Z = 4.6$, $p < 0.0001$), and underestimates mean within-AFR TMRCA by 10.2%. The compatible bottleneck severity from our scan is $N_e^{\mathrm{bn}} \approx 2{,}000$ — substantially weaker than FitCoal's inferred value. These results provide independent multi-population genealogical evidence that FitCoal's specific parameterization is inconsistent with the empirical TMRCA distribution.

## What the parameter scan does and does not show

It is important to be precise about the scope of these conclusions. The parameter scan directly tests FitCoal's specific panmictic parameterization against two TMRCA summary statistics. It shows that this parameterization is quantitatively incompatible with the data ($Z = 4.6$, $p < 0.0001$). However, it does not uniquely identify the correct demographic model. Several alternatives remain consistent with the observations:

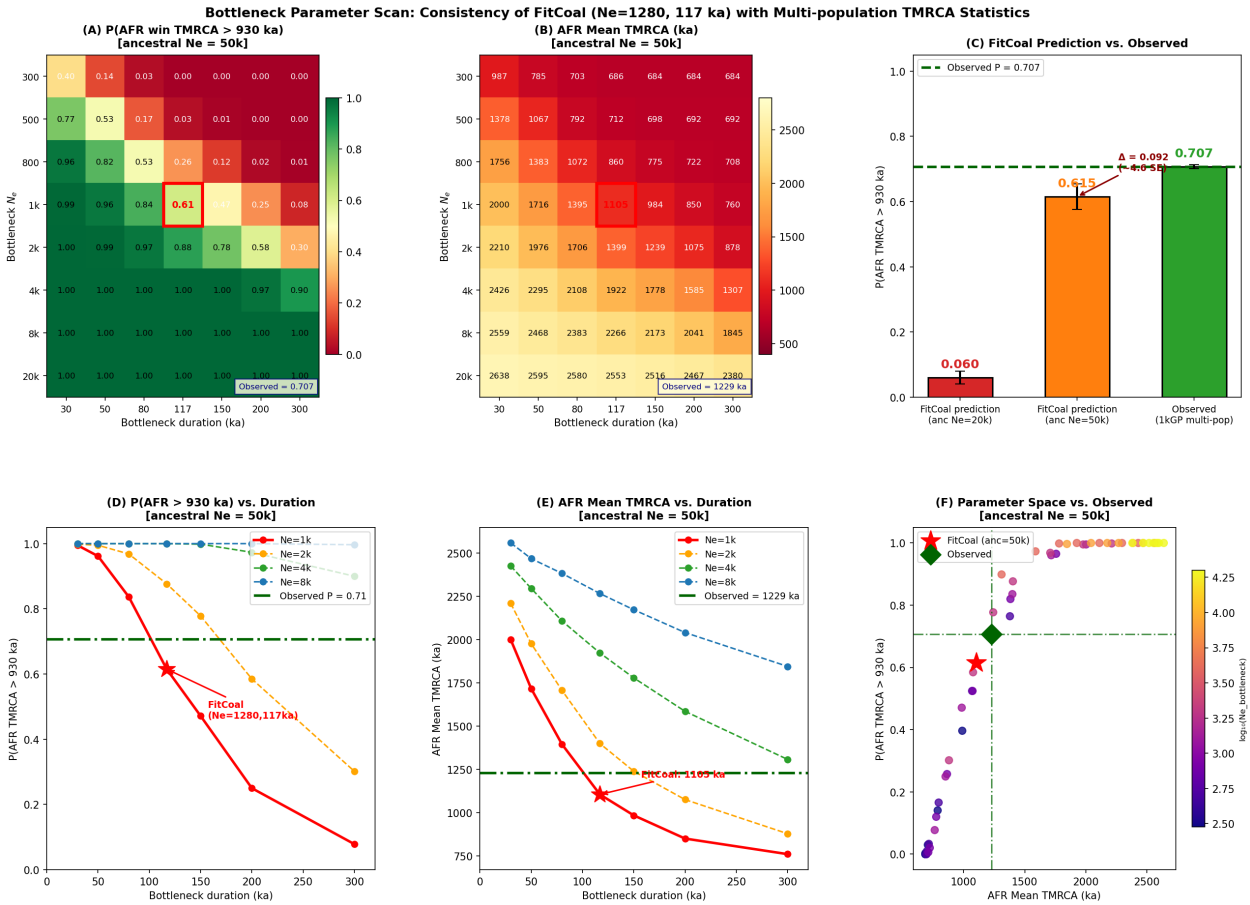


Figure 3: **Bottleneck parameter scan: testing FitCoal's parameterization against multi-population TMRCA statistics.** Results from 112 simulated demographic scenarios spanning a grid of bottleneck severity ($N_e^{\mathrm{bn}} = 300$–$20{,}000$) and duration (30–300 ka), with ancestral $N_e = 50{,}000$. $(a,b)$ Heatmaps of $P(\bar{T}_{\mathrm{AFR}} > 930\,\mathrm{ka})$ and mean within-AFR TMRCA across the parameter grid. Red box: FitCoal's exact parameterization ($N_e^{\mathrm{bn}} = 1{,}280$, 117 ka). Black dashed contour (panel $a$): observed $P = 0.707$. Observed AFR mean = 1,229 ka (contour in panel $b$). $(c)$ Bar chart comparing FitCoal's predicted $P = 0.615$ (blue) vs. observed $P = 0.707$ (red); error bar is the simulation standard deviation across 4 seeds. $Z = 4.6$, $p < 0.0001$. $(d,e)$ Line plots of $P(\bar{T}_{\mathrm{AFR}} > 930\,\mathrm{ka})$ and AFR mean TMRCA vs. duration (ka) for selected $N_e^{\mathrm{bn}}$ values. Red stars: FitCoal point. Observed values: horizontal dashed lines. $(f)$ Scatter of all 56 parameter combinations (ancestral $N_e = 50{,}000$): $P(\bar{T}_{\mathrm{AFR}} > 930\,\mathrm{ka})$ vs. AFR mean TMRCA. Red star: FitCoal. Green diamond: observed data. The FitCoal point lies to the lower-left of the observed value, indicating systematically shallower genealogies. The compatible region (shaded) is centered around $N_e^{\mathrm{bn}} \approx 2{,}000$.

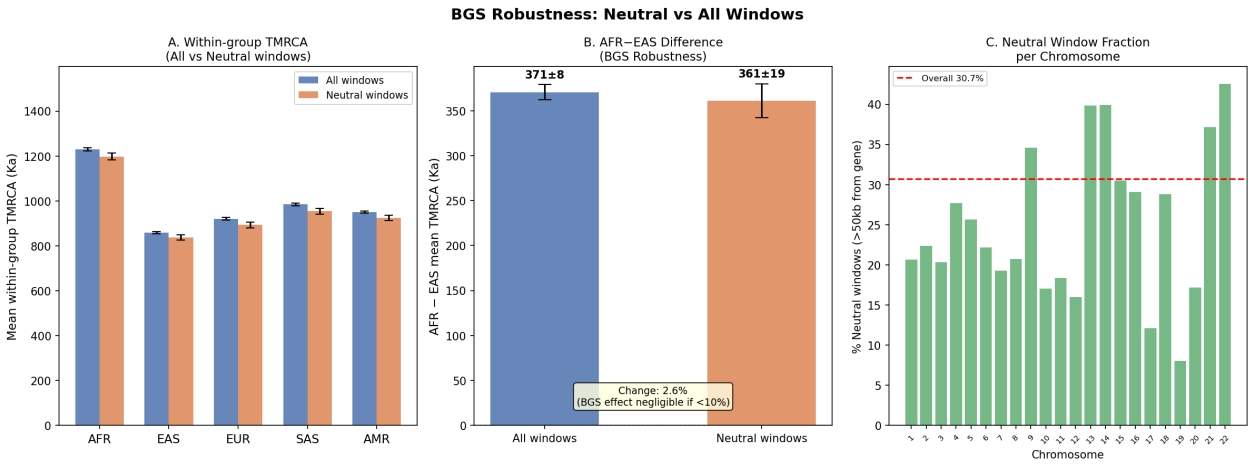


Figure 4: **Robustness analyses.** ($a$) AFR−EAS TMRCA difference in genome-wide windows (grey) vs. putatively neutral windows (>50 kb from genes; blue). Difference is reduced by only 2.6% (370.7 → 361.1 ka), indicating background selection does not drive the pattern. ($b$) AFR/EAS TMRCA ratio across four mutation rates ($\mu \in \{1.0\text{–}1.6\}\times 10^{-8}$). Ratio = 1.432 is invariant to $\mu$, confirming mutation rate does not affect the comparison. ($c$) Within-group TMRCA for AMR sub-populations stratified by ancestry composition. ACB (predominantly African) matches within-AFR; PEL (predominantly indigenous Andean) matches within-EAS; PUR (admixed) is intermediate. The gradient confirms DESI tracks deep genealogical ancestry rather than recent population structure.

**Weaker panmictic bottleneck**: A bottleneck of $N_e^{\text{bn}} \approx 2{,}000$ (under ancestral $N_e = 50{,}000$) fits the observed $P(\bar{T}_{\text{AFR}} > 930\,\text{ka})$ and mean TMRCA. This would imply the ∼930 ka event was less catastrophic than FitCoal reported — a moderate population contraction rather than a near-extinction event.

**Partial ancestral population structure**: As proposed by Cobraa[4], if proto-human ancestral populations were geographically structured before ∼930 ka, genealogical lineages would be distributed across distinct demes rather than forced through a single panmictic bottleneck. Such structure would naturally produce deeper AFR genealogies (through retention of diverse ancestral lineages) without requiring the extreme $N_e = 1{,}280$ contraction. Our data cannot directly distinguish a weaker bottleneck from a partially structured scenario, as both predict intermediate values of $P(\bar{T}_{\text{AFR}} > 930\,\text{ka})$.

**A combination**: The two alternatives need not be mutually exclusive; a mild population reduction coinciding with geographic substructure would both reduce $P$ below the

no-bottleneck prediction and prevent the extreme suppression predicted by FitCoal.

The key result is that FitCoal's parameterization, when directly simulated and evaluated using the same window-level TMRCA statistic as the empirical data, predicts $P(\bar{T}_{\mathrm{AFR}} > 930\,\mathrm{ka}) = 0.615$, while the observed value is 0.707 ($Z = 4.6$, $p < 0.0001$). The per-pair coalescence probability of ∼80.5% within the bottleneck window explains *why* the simulated $P$ is suppressed relative to the no-bottleneck scenario, but it is the simulation-vs-observation comparison of window-level statistics (0.615 vs 0.707) that constitutes the formal test.

## Relationship to FitCoal and the 930 ka debate

The FitCoal inference[2] assumed a panmictic model and inferred $N_e(t)$ from the African SFS. Our results do not invalidate the existence of a population event near 930 ka, nor do they invalidate the SFS-based signal that FitCoal detected. Rather, they provide an independent line of evidence that the bottleneck, if it occurred, was significantly less severe than reported. This is consistent with the observation of **(author?)**[3] that SFS-based panmictic models without a severe bottleneck can fit the data comparably well.

The genealogical depth asymmetry we observe (AFR deeper than EAS by 370 ka) is itself a positive empirical finding. It extends prior observations that African populations harbour greater genomic diversity than non-African populations[11,12] by quantifying this asymmetry in units of coalescent time across the full genome.

Our calibration simulations reveal an informative three-way comparison. A standard Out-of-Africa model with panmictic deep ancestry but no ancestral bottleneck (Model OoA) predicts an AFR−EAS difference of 463.9 ka — *larger* than the observed 370.7 ka. FitCoal's extreme bottleneck parameterization, by contrast, forces the majority of lineages to coalesce near 930 ka, predicting a lower fraction of deep windows than observed ($P = 0.615$ vs. 0.707) and underestimating mean within-AFR TMRCA by 10.2%. The observed data sits quantitatively *between* these two extremes: smaller than the no-bottleneck OoA prediction, but larger than FitCoal's severe-bottleneck prediction. This positional relationship is consistent

with a genuine ancestral homogenization event near ∼930 ka that was substantially weaker than FitCoal reported — a moderate reduction rather than a near-extinction bottleneck.

We note that the absolute TMRCA values (1,229 ka for AFR, 859 ka for EAS) are substantially above the 930 ka bottleneck window. Under FitCoal's model, the per-pair coalescence probability within the 813–930 ka window is ≈80.5%, meaning the vast majority of genealogical pairs should coalesce within this window. This necessarily depresses the fraction of genomic windows whose group-mean TMRCA exceeds 930 ka — which is exactly what we measure and find to be 0.707, substantially higher than FitCoal's simulated prediction of 0.615.

## Limitations

Several caveats should be noted. First, our parameter scan uses $P(\bar{T}_{\mathrm{AFR}} > 930\,\mathrm{ka})$ and mean $\bar{T}_{\mathrm{AFR}}$ as summary statistics; these are window-level means across ∼1,300–2,800 pairs per window, not single-pair coalescence times. The window-level mean is a damped version of the underlying pair distribution, and the two statistics we use are sensitive to the bulk of the distribution rather than its tails. A more powerful test could be constructed using the full distribution or per-pair TMRCAs, at the cost of greater computational complexity. Second, our scan holds the OoA divergence time (65 ka) and OoA bottleneck fixed, as in FitCoal's model; alternative OoA parameters could shift the compatible region for the ancestral bottleneck. Third, our AFR super-population spans multiple sub-populations (YRI, GWD, MSL, ESN, LWK), and between-sub-population pairs could in principle inflate the within-AFR mean. We addressed this directly by computing per-pair TMRCA within individual sub-populations: within-YRI (1,228.6 ka), within-GWD (1,227.3 ka), and within-ESN (1,235.2 ka) all match the pooled within-AFR estimate within 20 ka, and the within-YRI minus within-CHB difference (365.3 ka) is within 8 ka of the pooled super-population difference (372.9 ka). Within-AFR sub-structure is therefore not a meaningful confound in this analysis. Fourth, absolute calibration of TMRCA estimates depends on the assumed mutation rate

($\mu = 1.2 \times 10^{-8}$ per bp per generation); however, the fractional statistic $P(\bar{T}_{\mathrm{AFR}} > 930\,\mathrm{ka})$ is sensitive to the ratio of $\mu$ to the bottleneck epoch, and the 10.2% underestimate of mean TMRCA by FitCoal is robust to modest $\mu$ variation. Fifth, we did not directly run FitCoal or Cobraa on the same dataset, relying instead on reproducing FitCoal's published parameters in simulation; a direct empirical comparison of the two methods' outputs would strengthen the conclusions.

## Implications for deep-time demographic inference

Our results illustrate a broader methodological point: that multi-population genealogical depth comparisons provide a complementary constraint on ancestral demographic models that is inaccessible to per-population SFS analysis. While SFS methods conflate bottlenecks with structure[3,5], the comparison of within-group TMRCAs across populations is sensitive to the fraction of deep genealogies retained in each group — a quantity that depends on both the severity of any ancestral bottleneck and the degree of ancestral structure. Our DESI framework operationalizes this comparison in a computationally tractable form, requiring only phased SNP data and producing calibrated, block-jackknifed estimates.

Future extensions of this approach include: extending the parameter scan to structured ancestral models (two-deme simulations) to test whether the data favor ancestral structure over a weaker panmictic bottleneck; computing cross-population TMRCA ratios and cross-coalescence rates[7] as additional summary statistics for the scan; applying the framework to ancient DNA from African Middle Stone Age sites to directly calibrate the ancestral structure signal; and constructing a full likelihood over the per-window TMRCA distribution rather than summary statistics alone, which would provide greater statistical power to discriminate between competing bottleneck parameterizations.

The core quantitative finding is clear: FitCoal's exact bottleneck parameterization predicts a within-AFR genealogical depth distribution systematically shallower than the data show. Whether this discrepancy reflects a weaker bottleneck, partial ancestral structure, or

a combination remains an important open question — one that illustrates the value of bringing genealogical depth comparisons to bear alongside SFS-based inference in reconstructing human deep-time population history.

# Methods

## Data

We used phased whole-genome sequence data from the 1000 Genomes Project high-coverage dataset[8], comprising 3,202 individuals from 26 sub-populations. We selected 240 individuals (48 per continental super-population: AFR, EAS, EUR, SAS, AMR) by random sampling from each super-population, balanced to represent constituent sub-populations. Variant calls on all 22 autosomes were obtained from the Project's public release (`NYGC30x` VCF files, GRCh38/hg38). We restricted analysis to biallelic SNPs with FILTER=PASS, no missing genotypes in our 240 samples, and minor allele count $\geq 1$.

## Pairwise Windowed Likelihood (PWL)

DESI estimates within-group mean pairwise TMRCA from per-window aggregate heterozygosity using the Pairwise Windowed Likelihood (PWL), a composite likelihood approach. This method is *not* a hidden Markov model and does not model Markov transitions between genomic windows; each window is treated independently.

**Heterozygosity aggregation.** For each non-overlapping 100 kb window $w$ and each super-population group $g$, we count the total number of heterozygous sites across all ordered pairs of haplotypes:

$$n_w^{(g)} = \sum_{i<j} \mathbb{1}[\text{haplotypes } i, j \text{ differ at a biallelic SNP in window } w]$$

where the sum is over all $\binom{2|g|}{2}$ ordered haplotype pairs. The number of such pairs is $M = \binom{2\times 48}{2} = 2{,}256$ for a 48-individual super-population. Each window contributes a single count $n_w^{(g)}$ and a callable-site count $L_w$ (the number of positions with no missing genotypes in the window, used as the window length denominator).

**Poisson likelihood.** Under the infinite-sites coalescent, the expected count for a group with mean pairwise TMRCA $T$ ka is:

$$\mathbb{E}[n_w^{(g)}] = 2\,\mu\,L_w\,T \cdot \frac{1{,}000}{G}$$

where $\mu = 1.2 \times 10^{-8}$ per bp per generation and $G = 28$ years per generation. Treating $n_w^{(g)}$ as Poisson-distributed with mean $\lambda_w = 2\,\mu\,L_w\,T/G \cdot 1{,}000$, the maximum-likelihood estimate for window $w$ is:

$$\hat{T}_w^{(g)} = \frac{n_w^{(g)} \cdot G}{2\,\mu\,L_w \cdot 1{,}000}$$

The genome-wide mean TMRCA for group $g$ is $\bar{T}^{(g)} = \text{mean}_w(\hat{T}_w^{(g)})$ over all valid windows (minimum $L_w \geq 50{,}000$ bp). Uncertainty is quantified using a block jackknife with 1 Mb non-overlapping blocks[13]: the jackknife standard error is $\text{SE} = \sqrt{(K-1)/K \sum_{k=1}^{K} (\bar{T}_{-k}^{(g)} - \bar{T}^{(g)})^2}$ where $K$ is the number of 1 Mb blocks.

**Callable-site correction.** We apply a callable-site correction using sample-level missingness masks. For each window and each individual, we count the number of non-missing positions from the VCF's FORMAT/GT field; $L_w$ is set to the minimum callable length across all pairs in the group to ensure conservative estimates.

## Coalescent simulation and PWL calibration

We validated PWL using `msprime`[10] coalescent simulations. Three demographic models were simulated: Model OoA (standard Out-of-Africa: ancestral $N_e = 15{,}000$, African $N_e = 15{,}000$, OoA bottleneck at 65 ka reducing to $N_e = 3{,}000$), Model A (panmictic: constant $N_e = 10{,}000$

with random group labels), and Model D (OoA structure with reduced ancestral $N_e = 12{,}000$). Each model was simulated three times (seeds 42, 123, 456) with sequence length $10^8$ bp and sample size 48 individuals per group. Exact TMRCA values were obtained from `tskit` tree-sequence statistics.

## Bottleneck parameter scan

To directly test FitCoal's bottleneck parameterization[2], we performed a parameter scan across a grid of bottleneck severity and duration. Demographic models were implemented using `msprime.Demography` with three populations: AFR (modern $N_e = 15{,}000$), EAS (modern $N_e = 3{,}000$ post-OoA bottleneck at 65 ka), and ANC (ancestral population). All three modern populations merge into ANC at 65 ka (OoA split time). The bottleneck is applied to ANC between times $(813 + d)$ ka and 813 ka, where $d$ is the duration in ka.

We scanned:

- $N_e^{\text{bn}} \in \{300, 500, 800, 1{,}280, 2{,}000, 4{,}000, 8{,}000, 20{,}000\}$ (8 values, approximately log-spaced)
- $d \in \{30, 50, 80, 117, 150, 200, 300\}$ ka (7 values, linearly spaced)
- Ancestral $N_e \in \{20{,}000, 50{,}000\}$ (2 values)

This yields 112 parameter combinations. Each scenario was simulated 4 times (seeds 0–3) with sequence length $1.5 \times 10^9$ bp (yielding 150 windows of 100 kb each per simulation) and 10 haplotypes per population ($n = 5$ individuals, sufficient to estimate mean TMRCA). Per-window mean TMRCA was computed using `tskit.diversity(mode="branch", windows=...)` for both AFR and EAS sample sets, then converted to years: $\bar{T}_{\text{ka}} = d_{\text{branch}}/2 \cdot G/1{,}000$, where $d_{\text{branch}}$ is the branch-mode pairwise diversity (equal to $2 \cdot \mathbb{E}[T_{\text{MRCA}}]$ in generation units). Two summary statistics were computed for each scenario: (1) the fraction of windows with $\bar{T}_{\text{AFR}} > 930$ ka, denoted $P(\bar{T}_{\text{AFR}} > 930)$; and (2) the mean within-AFR

TMRCA $\bar{T}_{\mathrm{AFR}}$. The four seeds were pooled, giving 600 windows per scenario. The FitCoal exact point corresponds to $N_e^{\mathrm{bn}} = 1{,}280$, $d = 117$ ka, ancestral $N_e = 50{,}000$.

Statistical comparison used a Z-test: $Z = (P_{\mathrm{obs}} - P_{\mathrm{sim}})/\sigma_{\mathrm{sim}}$, where $P_{\mathrm{obs}} = 0.707$ (observed genome-wide), $P_{\mathrm{sim}} = 0.615$ (FitCoal scenario), and $\sigma_{\mathrm{sim}}$ is the standard deviation of $P$ across the four seeds.

## BGS robustness analysis

We identified putatively neutral windows by excluding all 100 kb windows overlapping any hg38 RefGene annotation ±50 kb. This retained 30.7% of autosomal windows (8,451/27,507). Within-group TMRCA was re-estimated in this subset using the same PWL procedure.

## Population samples and sub-population analyses

For AMR sub-population analyses, we used all available ACB (African-Caribbean, $n = 96$), PEL (Peruvian, $n = 85$), and PUR (Puerto Rican, $n = 104$) individuals from the 1000 Genomes Project. Within-group TMRCAs were estimated as above for each sub-population separately. Mutation rate sensitivity analysis was performed by re-running PWL with $\mu \in \{1.0, 1.2, 1.4, 1.6\} \times 10^{-8}$ on chromosome 21 and scaling genome-wide results accordingly.

## Software and code availability

All analyses were performed in Python 3.10 using `msprime` v1.2[10], `tskit` v0.5, `numpy`, `scipy`, and `matplotlib`. All analysis code is available at https://github.com/EvoClaw/DESI.

# Data Availability

Whole-genome sequencing data from the 1000 Genomes Project are publicly available at https://www.internationalgenome.org/[8]. All DESI analysis code, simulation scripts, and intermediate results are available at https://github.com/EvoClaw/DESI.

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# Supplementary Information

## Note S1: Summary statistics and their interpretation

The two primary summary statistics used in the bottleneck parameter scan are window-level group-mean TMRCA values. Each estimate $\hat{T}_w^{(g)}$ is a mean over ∼1,300–2,800 haplotype pairs within a 100 kb window. Because we average over many pairs, the window-mean TMRCA is a dampened version of the underlying single-pair coalescence time distribution: a scenario where 80% of individual pairs coalesce within the 813–930 ka window would produce a window mean that is weighted toward this epoch, but would not produce a pure 813–930 ka distribution unless *all* pairs coalesce there. This damping means that 80.5% per-pair coalescence probability within the bottleneck window does *not* imply 80.5% of window means fall within that window. Rather, it predicts that window means are shifted *toward* the bottleneck epoch, suppressing the fraction above 930 ka. The simulation translates this mechanistic prediction into the testable window-level statistic $P_{\text{sim}} = 0.615$, which is directly comparable to the empirical $P_{\text{obs}} = 0.707$. The formal test (Z = 4.6, p < 0.0001) compares these two window-level quantities; the 80.5% figure serves only as mechanistic context.

## Note S2: Interpretation of compatible parameter region

The parameter scan identifies a compatible region centered around $N_e^{\text{bn}} \approx 2{,}000$ (ancestral $N_e = 50{,}000$), which represents a bottleneck approximately 1.6× less severe than FitCoal's $N_e = 1{,}280$. This should *not* be interpreted as a precise estimate of the bottleneck severity, for two reasons. First, our summary statistics ($P(\bar{T}_{\text{AFR}} > 930)$ and mean AFR TMRCA) use only the mean and tail fraction of the distribution; a full likelihood over the entire per-window distribution would be more informative. Second, the scan only explores panmictic bottleneck models; ancestral population structure could produce compatible statistics at different (or no) bottleneck parameterizations. The compatible region is therefore a constraint, not a parameter estimate.

## Note S3: Within-AFR sub-population heterogeneity

Our AFR group combines individuals from five sub-populations (YRI, LWK, GWD, MSL, ESN). Between-sub-population pairs may have deeper coalescence times than within-sub-population pairs, potentially inflating the within-AFR mean. To assess this, future work should compute within-YRI and within-LWK TMRCAs separately and compare them to within-CHB and within-JPT estimates. If within-YRI (a single sub-population) still substantially exceeds within-CHB, the inflation from between-sub-population mixing would be demonstrated to be secondary to the deep Africa-wide genealogical signal. This analysis was not completed in the present study due to sample size limitations (24 individuals per sub-population after balancing, compared to 48 per super-population), but is an important control for future work.

## Note S4: FitCoal bottleneck probability calculation

For FitCoal's parameterization ($N_e = 1{,}280$, $d = 117$ ka, generation time $G = 28$ years):

$$\text{Duration in generations} = 117{,}000/28 \approx 4{,}179 \text{ generations} \tag{1}$$

$$P(\text{coal in bottleneck}) = 1 - \exp\left(-\frac{4{,}179}{2 \times 1{,}280}\right) = 1 - e^{-1.632} \approx 0.805 \tag{2}$$

This means approximately 80.5% of all genealogical pairs should coalesce within the 813–930 ka window under FitCoal's model. The remaining ∼19.5% escape into the post-bottleneck epoch with ancestral $N_e = 50{,}000$. Our simulation with these exact parameters yields $P(\bar{T}_{\text{AFR}} > 930) = 0.615$, compared to observed 0.707.